

Integration of artificial intelligence activities in software development processes and measuring effectiveness of integration

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Abstract: Recently, the modelling of whole process of software (SW) development is performed using extended waterfall and agile models. The further advancement of extended waterfall and agile models in the main phases like communication, planning, modelling, construction and deployment can improve the overall quality of the product. Accordingly, in this study, artificial intelligence (AI) activities are integrated into SW development processes. The important AI activities like intelligent agents, machine learning (ML), knowledge representation, statistical model, probabilistic methods, and fuzzy are integrated into the extended waterfall model. Again, AI activities like intelligent decision making, ML, Turing test, search and optimisation are integrated into the agile model. Two metrics such as, Usability Goals Achievement Metric and Index of Integration are evaluated in five independent SW projects. Once SW projects are developed using these models, feedback queries have been collected formally and the collected data are extensively analysed to identify the individual characteristics of products, identifying correlation behaviour of products with respect to model and metrics.

1 Introduction

In recent years, there is a remarkable growth in research in software (SW) development field. A well-known publication on waterfall scheme was presented by Winston Royce's in the year 1970 [1]. Waterfall scheme is the method that made a revolution in SW development field, where the SW development is considered to be tremendously complex and user-specific and several limitations may occur when it is not properly arranged. The waterfall scheme acts as the traditional way for SW development. In waterfall model, communication, planning, modelling, construction and deployment are the important stages. Moreover, this model consists of iterations in the form of system development life cycle phases. The waterfall model provides a lot of advantages to the SW developers. The first advantage is that the staged development cycle enforces control since a phase will certainly have a starting and ending, which permits the salesperson and the clients to set targets to decide the developments in a significant way. As the requirements and design are defined before the execution of code, the effort to be applied and the time will be minimised, hence, the divergence from the schedule is avoided.

Attaining the requirements and design at the early stage provides many advantages including quality enhancement. Furthermore, the faults that are expected to happen can be revealed and corrected in the design stage itself. In the testing phase, all the elements will be integrated and thus detecting the faults and correcting it will be complex and hence it is not performed. The starting two phases results in a proper specification and so, the purpose of waterfall model is to aid in expressing the knowledge to the team members dispersed in several locations. The application of waterfall process is possible only if the entire requirements presents for re-engineering the currently available SW. This model cannot be used for business purposes due to the existence of iterations between the phases. Furthermore, it requires more time and cost for the system development and implementation. If the requirements of the customers are uncertain, then definitely it will results in failure. Collecting the whole requirement details is extremely not possible in the initial stage of the project and this model works only after providing all the requirements [2].

In this paper, essential artificial intelligence (AI) activities, methods, deliverables and skills are identified and integrated in SW engineering (SE) processes. This work is about to integrate AI activities and proposes extended waterfall and agile models. This approach is to integrate all the essential AI activities with a point in the SE process where the AI deliverables will be most useful, and at the same time to keep the original intents of the SE processes intact. The integrated AI activities in extended waterfall and agile models are then evaluated using two metrics such as Usability Goals Achievement Metric (UGAM) and Index of Integration (IoI) which are used to demonstrate the impact of integrating AI activities. The first metric UGAM is a product metric that measures the extent to which the design of a product achieves its user-experience goals. The second metric IoI is a process metric that measures the extent of integration of the AI activities in the SE process. The paper is organised as follows: Section 2 presents the literature review and Section 3 presents the AI activities. Section 4 presents integration of AI activities in extended waterfall model and Section 5 presents the integration of AI activities in extended agile model. Section 6 presents the experimental study and Section 7 concludes the paper.

2 Literature review

In the literature, the discussions on various phases involved in SE like communication, modelling, planning, construction and deployment [3] can be seen. Hutchinson *et al.* [4] have dealt with a development process containing four stages, which depends on the components. The implementation of this model was naturally found to be more complicated. Instead of employing in-house development, the integration of the off-the-shelf components with the freshly developed components was the central theme behind this model and it failed to make use of the repository. Costabile [5] has discussed the means with which the human-computer interaction (HCI) activities and the waterfall model can be integrated. Göransson *et al.* [6] have come up with a novel discipline, termed as the usability design, to achieve integration between the HCI activities and the rational unified process. Joshi

and Sarda [7] have exploited IoI for integrating the HCI activities and the waterfall model. They advised to involve the investigation of the competitive product analysis, contextual user studies and user modelling, product definition/information architecture/wireframes with a multi-disciplinary team, ideation with a multi-disciplinary team, enrichment of product description and formative usability assessment in the communication phase. In addition, they insisted to make an inclusion of comprehensive user interface prototyping, enrichment of prototype and formative usability assessment of the user interface in the modelling phase of the waterfall model. At the time of construction phase, the usability team should perform re-assessments and aid the development team. Moreover, a summative usability assessment should be performed, if a fairly large fidelity version is found to be present.

Jain *et al.* [8] described that a SW project will never succeed, if the collected SW requirements are not much significant. The reason behind this failure is normally due to the shortcomings encountered in the approaches used for deciding the requirements. They put forward a method with which the SW can be built through the classification of components depending on the knowledge base. The previously available components were detected, opted and integrated within a newly emerged SW with the help of the knowledge base. In addition, they have given a description on the various categorising methodologies that allows the artefacts to be used again. These include keyword, enumerated, hypertext and faceted. This paper has mainly attempted to achieve the goal of making the requirement collection in an uncomplicated way. Yet, no complete model for component-based development (CBD) was proposed. de Almeida *et al.* [9] have introduced an incremental approach for distributed CBD, in which the working relies on two phases. In the initial phase, the requirements pertaining to the problem under consideration were dealt and the construction of the usable components with object oriented (OO) language was then accomplished. A repository was utilised to save these usable components. However, in the secondary phase, the desired components for developing the SW system were chosen with the MVCASE tool into which the SW engineers have searched.

Capetz *et al.* [10] have put effort to develop a SW life cycle, which was capable of assisting CBD via OO language. This model has enclosed three important phases namely, domain engineering, system analysis, design and implementation. Further, this model was posed with a serious challenge in choosing the reusable components, while the design phase was carried out. However, in actual case, the reusable components are chosen in the analysis phase because the details regarding the expense, agenda and the attempts associated with the development and integration of the components can be made known to the analyst for assessing them. Ning Jim [11] has presented a CBD model constituting of phases like, component analysis, component brokerage, architectural design, component integration and component production. In addition, the waterfall model was altered through incorporating the aforesaid phases in the CBD model. The commercial applications cannot use the waterfall model due to the repetitive checking of the phases. However, research projects alone can be applied with the waterfall process, demanding increased duration and expense.

In [12], prototype model has been introduced with the aim of integrating other approaches (for instance, waterfall approach) into it, so that the outcomes can be improved. In the prototype model, listening to the customer, developing a demo/rebuild and evaluating serves as the important phases. This model was a recurring model in which the iteration never ceases until the project gets finished. The development results were yielded in a rapid manner with improved cost-efficiency. The quality of SW can be degraded with the results gained quickly. The prototype model does not imply a detached model for developing the systems, but have greater dependence on the end user in the form of interfaces as well as the activities. Hence, the efficiency gets lowered. The fear about sustaining their job avoids the end user from showing interest on achieving a triumphant SW project and thus, few of the traits or functions in SW gets neglected. Adolph *et al.* [13] have revealed a grounded theory to show the way the

people can have control over the processes involved in developing the SW. The engineers engaged in developing the SW concentrates more on how the task can be fulfilled. The task completion is possible with a process containing four stages of reconciling perspectives. With reconciling Perspectives, the viewpoints of several people regarding a SW project can be satisfied. In addition, the individuals are asked to protect them from any undesired experiences from the surrounding to complete the SW project in an effective way.

Joshi *et al.* [14] have recommended two metrics for describing the influence of integrating HCI activities and SE processes. UGAM refers to a product metric, which gives the measure of the degree to which the product design satisfies the objectives of user skills. On the other hand, IoI points to a process metric that is capable of measuring the level to which the HCI activities and the SE processes get integrated. These two metrics have been produced from the standpoint of an organisation and hence, they can find applications in a vast number of projects or products. In addition, a trial on how the metrics can be utilised without difficulty in an industrial environment has been dealt. Since the two metrics have higher correlation to each other and allows a successful integration of HCI and SE processes, plenty of other methods can also use them in an identical sense. The two metrics were evaluated in three self-governing studies and they are, a classroom-dependent evaluation consisting of two sets of students, a qualitative feedback from three industrial projects and a quantitative evaluation with 61 industrial projects. These two metrics allow the process to be more organised, since they are beneficial and less complex.

Boehm [15] states the evolutionary model as a 'model whose stages consist of expanding increments of an operational SW product, with the direction of evolution being determined by operational experience'. Hence, the customary phases in the waterfall model are never planned to construct the entire system in the evolutionary approach. In contrast, the phases should be rapidly implemented plenty of times to create several product versions. In addition, the product should be constantly operated based on the feedback acquired and the never-ending process applied on the product should be executed through the addition of portions in an increasing manner or with the enhancement of the previously executed ones. Clarke and O'Connor [16] have merged a wide range of relevant research works with the primary reference structure of the environmental features, which has great impact on the SW development process. Precise data coding methods from grounded theory were employed to aid the consolidation of data and it was supposed that the framework gained would play a vital role in the knowledge domain of SE. Jalote *et al.* [17] have created the time boxing process model, which merges the notion behind the time boxed iterations as well as pipelining to carry out several iterations in an overlapping manner. If the time boxing process model is considered, every single time boxed iteration is found to get split into stages of same length. The stages, in turn, own a preset function and produce a better product that is offered to the subsequent stage later. Inclusion of the pipelining idea enables several time boxes to be implemented in a simultaneous fashion and hence, the product can be released and delivered within a shorter duration.

3 AI activities

AI technology is progressively more common in life today. It has applications in several industries including game theory and strategic planning, media, image processing, finance, and in the modern research areas such as robotics, natural language processing, medical diagnosis, and quantum science and so on. The aim of this paper is to use AI techniques in SW development processes for improving its performance. The major AI or machine learning (ML) activities are given below:

Intelligent agents [18]: An intelligent agent is a SW that performs by their own and aids people. Intelligent agent allows people to

handover their works to the agent SW. Agents can carry out repetitive tasks, remember the things forgotten by peoples, summarise complex data intelligently, learn from people and even provide recommendations to people.

ML [19]: It gives the capability to computers to learn without being programmed explicitly. In past years, ML has provided self-driving cars, effective web search, practical speech recognition, and immensely enhanced understanding of the human genome. ML is very common in day-to-day life and peoples use it several times in a day without knowing about it. Several researchers found it to be the most excellent way to develop towards human-level AI.

Knowledge representation [20]: It includes findings from psychology about how peoples solve problems and signifies knowledge to design formalisms that helps to design and build complex systems in easy way. Knowledge representation and reasoning also involves findings from logic to computerise several types of reasoning, such as the application of rules or the relations of sets and subsets.

Statistical model [21]: It represents a set of assumptions regarding the generation of the observed data and similar data from a larger population. A model signifies the data-generating process in significantly idealised form. The model assumption explains a set of probability distributions, a few of which are supposed to sufficiently approximate the distribution from which a specific data set is sampled.

Search and optimisation [22]: Searching is intended to find another choice which is much cost effective and achieve high performance with given constraints, and maximise the required factors and minimise the unwanted one. Maximisation indicates annoying to achieve the maximum result or outcome without considering the expense. Optimisation process is limited by missing of full information, and the time shortage to estimate the available information. In computer modelling of business problems, a linear programming technique of operation research is generally used for optimisation.

Probabilistic methods [23]: The probabilistic method is a significant technique to prove the availability of combinatorial objects with specific features. It is based on probability theory, but it can be used to prove theorems that do not use probability.

Turing test [24]: The Turing test is developed by Alan Turing in the year 1950. It is to test the capability of a machine to show intelligent behaviour equivalent to a human. Turing proposed that a human assessor would evaluate natural language conversations between a human and a machine that is designed to generate responses similar to human beings.

Fuzzy [25]: Fuzzy logic (FL) is an analytical control system that allows to implement itself in systems such as embedded micro-controllers, networked multi-channel PC or workstation-based data acquisition and control systems. It can be installed in hardware, SW, or in both hardware and SW combined together. FL presents an easy way to take an exact decision depending on unclear or missing input information.

3.1 AI activities to SW development processes

AI is a multi-disciplinary field with a focus on applying intelligence methods to several applications. AI emerged in 1980s with a focus on bringing automotive process in the usability of computer applications and productivity of users. With the spread of computing, AI researchers expanded interests to include areas such as social computing, ubiquitous computing, creativity, accessibility and entertainment. An intelligent agent which is an AI activity aims to deliver products that are useful, usable and desirable. Intelligent agents define 'the structure and behaviours of interactive products and services and user interactions with those products and services'. Following the fundamental tenets of user-centred design, the practice of AI-based design is grounded in an understanding of real users, their goals, tasks, experiences, needs and wants.

One view of waterfall or agile model is that it strives to develop high-quality SW. Are targeted users able to use the product? Do they need to be given a lot of training before they can start using it?

Does it take too long for users to complete tasks? Do users make too many errors while doing tasks? The answers to these questions can determine the difference between the success of a product and its failure. This is where the overlap of AI with waterfall or agile model becomes critical. In the early days, AI issues were limited as SW was deployed in few domains, on limited platforms, clients' requirements were clear and the SW product was often used internally by a few operators. Usability of products grew in importance as business success often depended on novice users being able to complete tasks. The last decade saw further expansion of the use of SW beyond productivity. In addition to usability, automatic formation determined the successes of products in the last decade and promises to do so in the next few years as well. The AI literature suggests many processes, activities, methods, skills and deliverables. A typical AI design process is made up of one or more phases, each of which may consist of one or more AI activities. The AI methods may result in a specific deliverable that may be an end in itself, or may be an input for another activity in the AI process or the SW development process.

4 Integrating AI activities in extended waterfall model

This section explains the enhanced waterfall model by modifying the extended waterfall model [26] using AI activities. The important AI methods explained in Section 3 are incorporated into the extended waterfall model. The extended waterfall model contains five important phases such as, communication, planning, construction, modelling and deployment for SW development processes. In the extended waterfall model, we have added some key concepts of AI in all the five phases of the waterfall model. Fig. 1 shows the enhanced waterfall model after integrating AI activities in extended waterfall model.

(a) *Communication:* In the communication phase of extended waterfall model, we have added the flexibility and synthetic process to improve the SW processes. The flexibility is added within the user studies using fuzzy logic, which is a flexible logic reasoning process. The user studies and its evaluation cannot be done with more accurate way due to the nature of different personalities involved. So, the flexible reasoning process is required to user studies and its evaluation. The flexible mechanism provided by the fuzzy sets can help the developers to perform automotive user studies and the evaluation is also easier due to the fuzzy inference process. Also, evaluation and refinement is an important step in the initial step for the SW development process. The evaluation of the SW at this stage can be done through synthetic data values which can be easily generated using the AI methods. This step can be very helpful to predict the future behaviour of the SW product even though the original data is not available. The imputation of the missing data and the synthetic data can be easily obtainable using different learning schemes which can generate the data as like the original data without any human intervention. This synthetic evaluation and refinement is also added in the communication phase.

(b) *Planning:* The second important step in the enhanced waterfall model is planning which is performed using estimation, scheduling and tracking. These processes are the common processes which are usually performed using traditional methods in most of the SW industries. Here, different AI methods like intelligence, optimisation and online process are applied to each and every process. The intelligence processes are fully automated process and it works like experts' brain so the planning may be more accurate. Also, the optimisation process yields the effective plan by searching through the candidate solution available in the space. Also, online handling of plan updates is also important to plan the SW processes more effectively on time. Accordingly, the intelligent estimation and the optimal scheduling and the online tracking are effectively integrated to get the improved performance in the SW development. The estimation of the outcome through the intelligent agents can lead the quality ensured products. Also, the optimal planning satisfies the

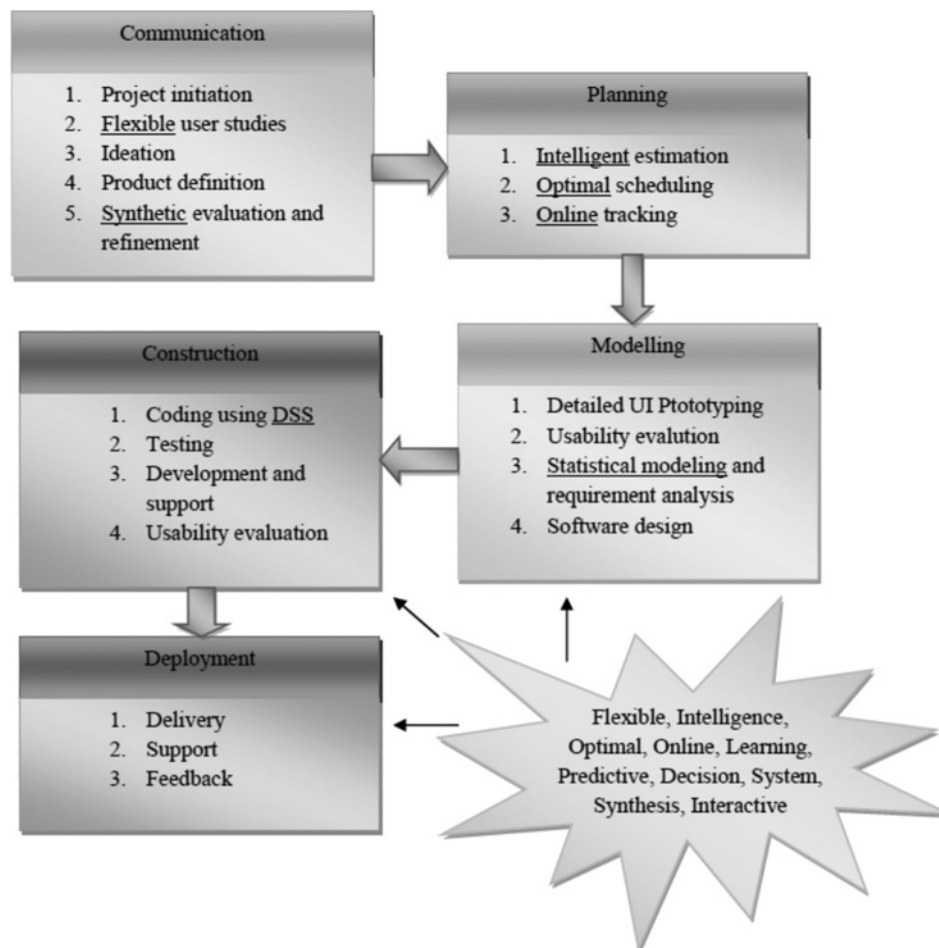


Fig. 1 Integrating AI activities in waterfall model

different constraints (connects start and goal states without collisions) and also optimises some quality metric.

(c) *Modelling*: Modelling is a third step to be performed in extended waterfall model for the SW development process. In the modelling phase, the important steps such as, UI prototyping, usability evaluation, requirement analysis and SW design are performed. Here, statistical modelling is additionally integrated to model the whole processes. The statistical modelling contains the probability model, regression modelling and mixture models which are the popular methods utilised in the AI. Statistical modelling is a simplified, mathematically-formalised way to approximate reality (i. e. what generates your data) and optionally to make predictions from this approximation. A statistical model is a class of mathematical model, which embodies a set of assumptions concerning the generation of some sample data, and similar data from a larger population. A statistical model represents, often in considerably idealised form, the data-generating process. As such: 'a model is a formal representation of a theory'. The reason of integrating the statistical modelling in the proposed waterfall model is that the modelled event through statistical features can be helpful to predict the outcome of the SW product from the initial step itself.

(d) *Construction*: The construction phase involves four activities in the waterfall model: coding, testing, development & support and usability evaluation. These are the core activities in the extended waterfall model. We do not suggest any changes to the last two phase's activities. However, we propose two activities that could have significant impact on the delivered SW: *Coding using DSS*: The coding of any SW products can be improved by integrating with the decision support system. The DSS can help the programmers to suggest or provide the required information to speed up the development phase. DSS analyses SW programs and presents it so that programmers can make decisions more easily. A

decision support system may present information graphically and may include an expert system or AI. It may be aimed at business executives or some other group of knowledge workers. The reason of integrating the decision support system is that it can reduce the human error in suggesting the required information and thereby reducing the debugging time. Turing testing: Testing of product can be further improved with Turing test which is much effective by integrating the intelligent behaviour.

(e) *Deployment*: We have not included any development related AI activities in the deployment phase of the extended waterfall model. The common processes in the deployment phase are delivery, support and feedback to give the exploitation for the users.

5 Integrating AI activities in agile model

This section explains the enhanced agile model by modifying the extended agile model given in [26] using the AI activities. The important AI methods explained in Section 3 are incorporated into the extended agile model to improve the SW development processes. The extended agile model contains two important phases such as, communication and iterative phases. The iterative phase contains the steps like, release, planning, design, test and coding. In the extended agile model, we have added some of the key concepts of AI in all the two phases of the agile model. Fig. 2 shows the enhanced agile model after integrating AI activities in extended agile model. The important steps involved in the extended agile model are as follows:

(1) *Communication*: The communication process contains five important steps like, user studied, ideation, product definition, usability evaluation. This communication phase includes the

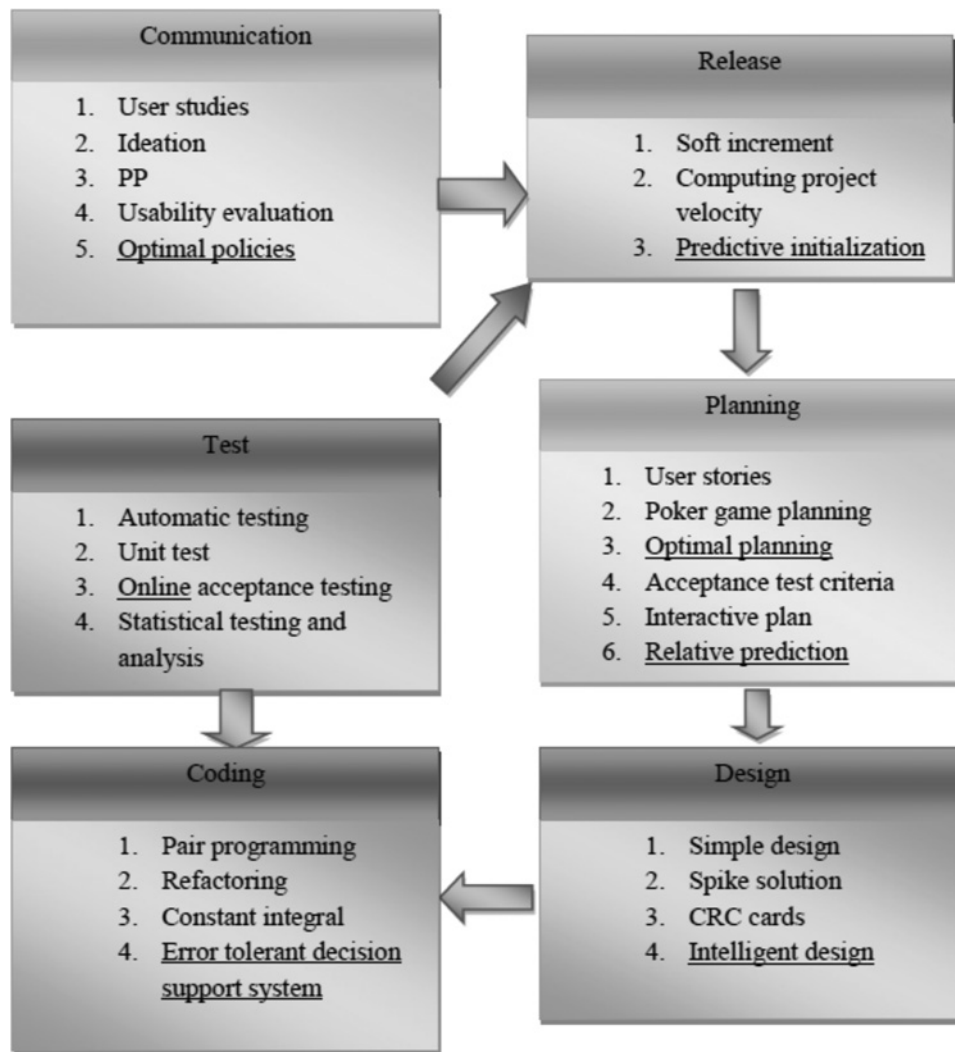


Fig. 2 Integrating AI activities in agile model

optimal policies to define the products and its evaluation criteria. The policies can be defined through an advanced natural language processes and optimisation algorithm. The intention of integrating these steps is to define the product definition with the policies formulated. The optimal policies are required on developing and implementing the SW products to associate for receiving alerts and information associated with distress situations, planning and coordinating operations, facility transits and searches, providing immediate assistance. Optimal policy is an important concept in decision.

(2) *Iterative phase*: The iterative phase includes five different steps which are operated in an iterative way until to get the desired results. Here, five steps such as, release, planning, design, coding and test are performed iteratively.

(i) *Release*: This step we additionally added predictive initialisation which is required to predict and compute the velocity of the project. The predictive models can be taken from the field of AI to improve the performance of the SW development cycle. Predictive modelling is the process of creating, testing and validating a model to best predict the probability of an outcome. A number of modelling methods from ML, artificial intelligence and statistics are available in predictive analytics SW solutions for this task. The model is chosen on the basis of testing, validation and evaluation using the detection theory to guess the probability of an outcome in a given set amount of input data. Models can use one or more classifiers in trying to determine the probability of a set of data belonging to another set.

(ii) *Planning*: The extended agile model contains user stories, poker game planning, acceptance criteria and interactive plan under the

phase of planning. Here, optimal planning is added newly in the planning phase because it can consider the different constraints and objectives to formulate the plan chart which will be helpful for respective team to find their tasks optimally without migration on any defined tasks. The optimal planning is one of the most important techniques widely studied in all the discipline for preparing the plans or schedule. An optimal plan for the development of a SW is drawn up on the basis of the solution of a problem in mathematical programming, in which the conditionally external value (maximum or minimum) of a function is found (the target function, or criterion of optimality, of a plan). A specific plan variant can be selected on the basis of the criterion of optimality and thus, optimal planning can be applied at all levels of the planning and the development cycle. Also, relative prediction is also integrated for predicting the outcome of the plan chart. This relative prediction based on AI methods can improve the plan chart. The intention of integrating the relative prediction is to reduce the iterative phase of agile model. (iii) *Design*: Design process consists of the simple design, spike solutions, CRC cards and prototype in the extended agile model. In this paper, intelligent design and interactive user interface can be included to improve the design process. Interactive design is defined as a user-oriented field of study that focuses on meaningful communication of media through cyclical and collaborative processes between people and technology. Successful interactive designs have simple, clearly defined goals, a strong purpose and intuitive screen interface. The nature of interactive design tools as well as its impact on humans, are studied extensively in the field of HCI which accepts input from humans as it runs. Also, the automatic suggestion tool can

Table 1 UGAM attributes

learnability
findability: options/data/information should be visible/easy to find
user should take less time to learn: (e.g. in <10 min, in <2 h practice, in <2nd attempt)
users should be able to learn on their own
product should be internally consistent
product should be consistent with other products, older methods/past habits of users
product should be consistent with the earlier version
user should remember/retain critical, but infrequent tasks
speed of use
user must be able to do the primary task/the most frequent tasks quickly, easily, at all times
user should be able to navigate quickly and easily
product should not load user's memory/product should not put cognitive load on a user
flexibility: user should control the sequence of tasks
user should be able to complete frequent/critical tasks in specific time/no. of steps/in less efforts
product should be personalised for the user automatically
product should be localised for specific market segments
users should be able to customise the product for themselves
ease of use
interface should clearly communicate the conceptual model
intuitiveness: User should be able to predict the next step/task
no entry barrier: user must be able to complete critical tasks
product should require no unnecessary tasks
product should automate routine tasks/minimise user task load
product should be always on, always accessible
ease of communication
information architecture: Information should be well aggregated, well categorised, well presented
communication should be clear/user should easily understand text, visuals
error-free use
product should give good feedback/display its current status
product should not induce errors 375 product should tolerate user's errors/forgiving interface/should prevent errors
product should help user recover from errors/help users troubleshoot problems
subjective satisfaction
user should feel in control of the product/behavioural appeal
user should feel emotionally engaged with product/brand/product should be fun/reflective appeal
user should find the product aesthetically appealing/product should have a visceral appeal

be integrated within the system to further improve the design process. If these important techniques are integrated within the SW development process, the manual cost can be reduced and also, the performance in terms of cost and one quality will be improved.

(iv) *Coding*: Along with the pair programming, refactoring, continuous integration defined in the extended agile model, error tolerant decision support system is effectively integrated for making easy to write a code without much human intervention and the human mistakes. Error-tolerant support system is one that does not unduly penalise user or human errors. It is the human equivalent of fault tolerant system that allows programs to continue functioning in the presence of errors and suggesting the possible ways of overcoming tips. This AI-based support system can reduce the programming time and improve the debugging phase in both the quality and times.

(v) *Test*: In this phase of enhanced agile model, objective and constraint aware optimal testing phases are integrated and it is validated using online procedures. In the optimal testing scenario, multi-objective based regression testing is used collectively in the optimisation algorithm to find the test suite by satisfying the test requirements with the objectives like, coverage, fault detection, capability and cost. The important activities to be performed under the test procedure are automatic testing, unit test, online acceptance testing and statistical testing and analysis.

6 Measuring the effectiveness of integration

The proposed waterfall and agile models are evaluated using five products such as, supermarket SW, hospital management system, weight monitoring (android application), scheduling of duties

(android application) and health prediction system. *Super market SW* is the provision of maintaining stocks/ledgers of multi-location warehouse. This system alerts of the stocks which are going to expiry and the out of stocks. Also, the billing facility is included within the SW. *The hospital management system* contains the important facilities like, hospital appointment scheduling by patients, secure storage of hospital records, appointment reminders and quick printable prescriptions and bills. *Weight monitoring SW* records the weights of person on daily basis and the trend of the weights are visualised to the user more graphically in various extracted information. *Scheduling SW* is especially designed to assist in the complex task of organising employee shift schedules for small organisations. The automated planner that can generate schedules based on employees' availability, days off, leave, events, bookings. *The health prediction system* allows users to get instant guidance on their health issues through an intelligent health care system online. The system is fed with various symptoms and the disease/illness associated with those systems. The system allows user to share their symptoms and issues. It then processes user's symptoms to check for various illnesses that could be associated with it.

The proposed two models are applied to all the five products from the communication phase to final delivery phase and the SW development performance is evaluated. To evaluate the performance of the model using IoI, ten programmers those who are doing their masters in computer science are selected. To evaluate UGAM score, ten students came from an electrical discipline are selected.

6.1 Metrics

The proposed SW development models are evaluated using two metrics like, UGAM and IoI [14]. UGAM is a popular metric used to measure the quality of SW using experience of the user. UGAM is product metric on a scale of 0–100, where 100 indicates the best user experience possible and 0 indicates the worst. UGAM contains the following conceptual elements: Goals: high-level user-experience goals direct the design of interactive systems. Goal parameters: each high-level user-experience goal is broken down into a set of parameters that assist a designer to attain and assess the achievement of the higher-level goal in a direct manner. Score: each goal parameter has a score between 0 and 100, where 0 indicates the worst possible user-experience design on account of that parameter and 100 indicates the best possible user-experience design rated against the goal parameter. Weight: each goal parameter has a weight between 0 and 5. Guidelines: the reason of the guidelines is to guide evaluators give a score to parameters

$$UGAM = \frac{\sum_{i=1}^m w_i * A_i}{\sum_{i=1}^m w_i}$$

where w_i is the weight of the goal parameter, A is the score value of the goal parameter and m is the number of goal parameters. The attributes belonging to UGAM is shown in Table 1. The team first allocated weights to each goal parameter. The evaluators then assigned a score for each goal parameter. The weighted average for the goal parameter scores under each goal resulted in the score of that goal and the weighted average of all goal parameter scores resulted in the UGAM score. A weighted average of only goal parameters under one high level goal gives the score against that goal.

The IoI metric contains the following conceptual elements: SE phases: these are the broad phases as mentioned in the SE process models as given. Weight: each HCI activity is measured a weight on the scale of 0–5 where 0 represents that the activity is not relevant and 5 represents that the activity is particularly significance in the context of the project. Score: each activity has a score relevant with it. The score is given on a rating of 0–100, where 100 indicates the best case situation, i.e. the activity was done in the best possible manner and 0 indicates the worst-case

Table 2 IoI attributes

communication
contextual user studies and user modelling, competitive product analysis
ideation with a multidisciplinary team (HCI, tech, biz)
product definition/information architecture/wireframes with a multidisciplinary team (HCI, tech, biz)
usability evaluation (formative) and refinement of product definition modelling
detailed UI prototyping
usability evaluation (formative) and refinement of prototype construction
development support reviews by usability team
usability evaluation (summative)

situation where the activity was not done at all. Activity evaluation guidelines: these spell out considerations that guide the evaluation of each activity (Table 2). Guidelines may define the techniques utilised to take out activities, the skill and experience levels of the people doing the activities, the deliverables and other parameters that concern the fidelity of the activity

$$IOI = \sum_{i=1}^m \left(w_i * A_i / \sum_{i=1}^m w_i \right)$$

where w_i is the weight of the goal parameter, A is the score value of the goal parameter and m is the number of goal parameters. In IoI calculation, the weights are assigned by the senior professional from the company and a group of reviewers comprising of some project insiders and outsiders reviewed and rated the activities based on the guidelines, and its IoI is calculated based on the formulae.

6.2 Sample UGAM and IoI score

The proposed waterfall and agile model are evaluated using IoI and UGAM score. The IoI metrics are related to the development process so it is given to the team of developer to obtain the attributes values. Similarity, UGAM metric is related with usability measurement so it is given to the users of the product to obtain the attributes. Here, ten different developer or users are taken and the attribute values relevant to UGAM and IoI are collected. Based on the attribute value, the UGAM and IoI are computed as per the formulae. The

sample UGAM and IoI metrics are given in Table 3 for the ten different developers or users. The computed values are ranging from 10 to 50.

6.3 Correlation among the products

This section describes the correlation of five different products based on the metrics and model. The correlation is computed based on Pearson correlation coefficient [27]:

Product 1: The correlation of product 1 with other products in the perspective of metrics and models are given in Table 4. From the table, we understand that products 2 and 3 are having negative correlation in terms of waterfall model with respect to IoI metrics. IoI metrics with agile model have good correlation among all the products. Product 1 has the correlation ratio of 0.81. 0.85, 0.88 and 0.83 with products 2–5. Also, UGAM-based evaluation shows that the product has the positive correlation among the products. This correlative analysis ensured that the IoI has a somewhat greater effect in projects that use the agile process model rather than the waterfall process model.

Product 2: Table 5 shows the correlation of product 2 with other products. This analysis showed that product 2 with product 1 has the negative correlation among the other correlative analysis. Here, products 2 and 3 have the good correlation of 0.97 in terms of IoI-based agile model. This study reveals that the performance of the two proposed models has good correlation among the different metrics except the IoI-based waterfall model because the correlation value is more than 0.7 for most of the cases.

Product 3: Table 6 shows the correlative analysis of product 3 with other four products. Here, IoI-based waterfall model shows the negative correlation with other products. However, IoI-agile, UGAM-waterfall and UGAM-agile methods show the positive correlation among the products. Here, a good correlation of 0.95 is achieved in between products 3 and 4 in terms of IoI-agile model. This study shows the UGAM-based models have good correlation than the IoI-based models.

Product 4: The correlation of product 4 with other products in the perspective of metrics and models are given in Table 7. From the table, we understand that products 4 and 3 are having negative correlation in terms of waterfall model with respect to IoI metrics. IoI metrics with agile model have good correlation among all the products by reaching the value of 0.96. Product 4 has the

Table 3 Sample UGAM and IoI score

	U1	U2	U3	U4	U5	U6	U7	U8	U9	U10
IoI-agile	29.3	38.3	20.9	45	19.6	41.8	28.2	31.4	27	46.7
IoI-waterfall	27.5	22.5	16.6	25.8	25	25.8	29.1	25	20	14.1
UGAM-agile	37	46.1	22.8	47.8	33.1	35.3	32.7	35.3	34	34.4
UGAM-waterfall	37.5	40.5	22.8	46.9	31.4	35.3	31.4	36.2	33.1	33.6

Table 4 Correlation of product 1

Metrics	Model	1,1	1,2	1,3	1,4	1,5
IOI	waterfall	1	−0.27431	−0.47259	0.209428	0.049363
IOI	agile	1	0.813204	0.856672	0.886191	0.8309
UGAM	waterfall	1	0.797268	0.737043	0.632011	0.562849
UGAM	agile	1	0.75752	0.650968	0.644008	0.640513

Table 5 Correlation of product 2

		2,1	2,2	2,3	2,4	2,5
IOI	waterfall	−0.27431	1	0.450829	0.518316	0.55485
IOI	agile	0.813204	1	0.972016	0.865044	0.897637
UGAM	waterfall	0.797268	1	0.907341	0.760854	0.713286
UGAM	agile	0.75752	1	0.807385	0.785985	0.728743

Table 6 Correlation of product 3

		3,1	3,2	3,3	3,4	3,5
IOI	waterfall	-0.47259	0.450829	1	-0.02465	-0.03582
IOI	agile	0.856672	0.972016	1	0.935551	0.907131
UGAM	waterfall	0.737043	0.907341	1	0.865957	0.893782
UGAM	agile	0.650968	0.807385	1	0.762868	0.895172

Table 7 Correlation of product 4

		4,1	4,2	4,3	4,4	4,5
IOI	waterfall	0.209428	0.518316	-0.02465	1	0.966193
IOI	agile	0.886191	0.865044	0.935551	1	0.843984
UGAM	waterfall	0.632011	0.760854	0.865957	1	0.941166
UGAM	agile	0.644008	0.785985	0.762868	1	0.892199

Table 8 Correlation of product 5

		5,1	5,2	5,3	5,4	5,5
IOI	waterfall	0.049363	0.55485	-0.03582	0.966193	1
IOI	agile	0.8309	0.897637	0.907131	0.843984	1
UGAM	waterfall	0.562849	0.713286	0.893782	0.941166	1
UGAM	agile	0.640513	0.728743	0.895172	0.892199	1

Table 9 Correlation analysis among the metrics

	P1	P2	P3	P4	P5
IoI	-0.11048	-0.25388	-0.3329	-0.0925	0.253709
UGAM	0.96851	0.993599	0.880013	0.973867	0.982235

correlation ratio of 0.88, 0.86, 0.93 and 0.84 with products 1–3 and 5 for IoI with agile model. This study reveals that the IoI-based agile model shows the greater impact among the products as compared with UGAM-based model.

Product 5: Table 8 shows the correlation of product 5 with other products. This analysis shows that product 5 with product 3 has the negative correlation among the other correlative analysis. Here, products 5 and 4 have the good correlation of 0.94 in terms of UGAM-based waterfall model. This study reveals that the performance of the two proposed models has good correlation among the different metrics except product 3 because the correlation value is more than 0.7 for most of the cases.

6.4 Correlative analysis among the metrics

This section shows the correlative analysis of IoI metric of agile and waterfall models and the UGAM metric of agile and waterfall models. IoI metrics of both the models are having negative correlation in all the five products except product 5 according to Table 9. This means that the waterfall and agile models have not much correlation in terms of IoI metrics. However, UGAM metrics have good correlation among both the models. It shows the good correlation of 0.99 as compared with IoI metrics. This ensured that the UGAM-based evaluation shows similar behaviour for both the agile and waterfall models.

Table 10 Correlation analysis among the models

	P1	P2	P3	P4	P5
enhanced waterfall model	0.406998	-0.07852	-0.55435	0.598076	0.613313
enhanced agile model	0.641066	0.794018	0.674658	0.387911	0.536469

6.5 Correlative analysis among the models

This section shows the correlative analysis among the two models using two metrics IoI and UGAM. Table 10 shows the correlative analysis among the models. From the table, we understand that waterfall model having negative correlation for both the metrics taken for study. However, the agile model has good correlation in terms of both the metrics. This ensured that the agile model influences the better performance among the programmers as well as user due to the significant work study. The agile model shows the good correlation of 0.79 as compared with waterfall model.

7 Conclusion

This paper aims to improve and plan AI activities in SW development processes. We have proposed an architecture to organise AI activities and use it as a baseline to extend SE process models. Accordingly, extended waterfall and agile models are enhanced by integrating AI activities. For the performance study, five products such as, supermarket SW, hospital management system, weight monitoring (android application), scheduling of duties (android application) and health prediction system are considered. The proposed two models are applied to all the five products from the communication phase to final delivery phase and the SW development performance is evaluated. For the evaluation, UGAM and IoI metrics are utilised to help and evaluate the product against those goals. By demonstrating the correlation between IoI and UGAM in industry projects, we have validated our process model extensions for waterfall and agile processes. From the correlative study, we proved that IoI metrics have good correlation than the UGAM for the enhanced waterfall model.

8 References

- 1 Royce, W.: 'Managing the development of large software systems: Concepts and techniques'. Western Electric show and Convention Technical Papers, 1970
- 2 Laplante Phillip, A., Neill Colin, J.: 'The demise of the waterfall model is imminent and other urban myths' (Queue ACM Press, 2004), vol. 1, no. 10
- 3 Pressman, R.: 'Software engineering – a practitioner's approach' (McGraw-Hill, 2005, 6th edn.)
- 4 Hutchinson, J., Kotonya, G., Sommerville, I., *et al.*: 'A service model for component-based development'. Proc 30th EUROMICRO Conf., 2004

- 5 Costabile, M.F.: 'Usability in the software life cycle', in Chang, S.K. (Ed.): 'Handbook of software engineering and knowledge engineering', vol. 1 (New Jersey, World Scientific, 2001), pp. 179–192
- 6 Göransson, B., Lif, M., Gulliksen, J.: 'Usability design – extending rational unified process with a new discipline'. Int. Workshop on Interactive Systems Design, Specification, and Verification, 2003
- 7 Joshi, A., Sarda, N.L.: 'HCI and SE: towards a 'truly' unified waterfall process'. HCI Int. '07, 2007
- 8 Hemant, J., Vitharana, P., Zahedi, F.M.: 'An assessment model for requirements identification in component-based software development', *ACM SIGMIS DATA BASE*, 2003, **34**, (4), pp. 48–63
- 9 de Almeida, E.S., Alvaro, A., Lucredio, D., *et al.*: 'Distributed component-based software development: an incremental approach'. Proc. 28th Annual Int. Computer Software and Applied Conf., 2004
- 10 Capetz, L.F., Capretz, M.A.M., Li, D.: 'Component based software development'. Proc. 27th Annual Conf. of the IEEE Industrial Electronics Society, 2001
- 11 Ning Jim, Q.: 'A component-based software development model'. IEEE Software, 1996, pp. 389–394
- 12 Purtilo, J., Larson, A., Clark, J.: 'A methodology for prototyping-in-the-large'. Proc. 13th Int. Conf. on Software Engineering, 1991
- 13 Adolph, S., Kruchten, P., Hall, W.: 'Reconciling perspectives: a grounded theory of how people manage the process of software development', *J. Syst. Softw.*, 2012, **85**, pp. 1269–1286
- 14 Joshi, A., Sarda, N.L., Tripathi, S.: 'Measuring effectiveness of HCI integration in software development processes', *J. Syst. Softw.*, 2010, **83**, pp. 2045–2058
- 15 Boehm, B.: 'A spiral model of software development and enhancement', *IEEE Comput.*, 1988, **21**, (15), pp. 61–72
- 16 Clarke, P., O'Connor, R.V.: 'The situational factors that affect the software development process: towards a comprehensive reference framework', *Inf. Softw. Technol.*, 2012, **54**, pp. 433–447
- 17 Jalote, P., Palit, A., Kurien, P.: 'The timeboxing process model for iterative software development', *J. Syst. Softw.*, 2004, **70**, (1–2), pp. 117–127
- 18 Russell, S.J., Norvig, P.: 'Artificial intelligence: a modern approach' (Prentice-Hall, Upper Saddle River, New Jersey, 2003, 2nd edn.), ISBN 0-13-790395-2
- 19 Hinton, G.E.: 'Learning multiple layers of representation', *Trends Cogn. Sci.*, 2007, **11**, pp. 428–434
- 20 Lenat, D., Guha, R.V.: 'Building large knowledge-based systems' (Addison-Wesley, 1989), ISBN 0-201-51752-3
- 21 Stigler, S.M.: 'A historical view of statistical concepts in psychology and educational research', *Am. J. Educ.*, 1992, **101**, (1), pp. 60–70
- 22 Akbari, Z.: 'A multilevel evolutionary algorithm for optimizing numerical functions', *IJIEC*, 2011, **2**, pp. 419–430
- 23 Srinivas, M., Patnaik, L.: 'Adaptive probabilities of crossover and mutation in genetic algorithms', *IEEE Trans. Syst. Man Cybern.*, 1994, **24**, (4), pp. 656–667
- 24 Turing, A.M.: 'Computing machinery and intelligence', *Mind LIX*, 1950, **59**, (236), pp. 433–460
- 25 Zadeh, L.A.: 'Fuzzy sets', *Inf. Control*, 1965, **8**, (3), pp. 338–353
- 26 Joshi, A.: 'HCI + SE integration – case studies from offshore development projects'. Proc. of the Workshop on Increasing the Impact of Usability Work in Software Development; CHI 2007, San Jose, 2007
- 27 Soper, H.E., Young, A.W., Cave, B.M., *et al.*: 'On the distribution of the correlation coefficient in small samples. Appendix II to the papers of 'Student' and R.A. Fisher. A co-operative study', *Biometrika*, 1917, **11**, pp. 328–413